International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2023-0014
© 2023 Human Kinetics, Inc.
First Published Online: Oct. 10, 2023

Human Kinetics
ORIGINAL INVESTIGATION

Match Running Performance in Australian Football Is Related to Muscle Fiber Typology

Henry J. Hopwood,[1,2] Phillip M. Bellinger,[1] Heidi R. Compton,[2,3] Matthew N. Bourne,[1] Wim Derave,[4] Eline Lievens,[4] Ben Kennedy,[5] and Clare L. Minahan[1,6]

[1]Griffith Sport Science, Griffith University, Gold Coast, QLD, Australia; [2]Football Department, Gold Coast Football Club, Gold Coast, QLD, Australia; [3]Applied Sports Science and Exercise Testing Laboratory, University of Newcastle, Ourimbah, NSW, Australia; [4]Department of Movement and Sport Sciences, Ghent University, Ghent, Belgium; [5]Mermaid Beach Radiology, Gold Coast, QLD, Australia; [6]Australian Institute of Sport, Australian Sports Commission, Canberra, QLD, Australia

Purpose: To examine the association between muscle fiber typology and match running performance in professional Australian football (AF) athletes. ***Methods:*** An observational time–motion analysis was performed on 23 professional AF athletes during 224 games throughout the 2020 competitive season. Athletes were categorized by position as hybrid, small, or tall. Athlete running performance was measured using Global Navigation Satellite System devices. Mean total match running performance and maximal mean intensity values were calculated for moving mean durations between 1 and 10 minutes for speed (in meters per minute), high-speed-running distance (HSR, >4.17 $m \cdot s^{-1}$), and acceleration (in meters per second squared), while intercept and slopes were calculated using power law. Carnosine content was quantified by proton magnetic resonance spectroscopy in the gastrocnemius and soleus and expressed as a carnosine aggregate z score (CAZ score) to estimate muscle fiber typology. Mixed linear models were used to determine the association between CAZ score and running performance. ***Results:*** The mean (range) CAZ score was –0.60 (–1.89 to 1.25), indicating that most athletes possessed a greater estimated proportion of type I muscle fibers. A greater estimated proportion of type I fibers (ie, higher CAZ score) was associated with a larger accumulation of HSR (>4.17 $m \cdot s^{-1}$) and an increased ability to maintain HSR as the peak period duration increased. ***Conclusion:*** AF athletes with a greater estimated proportion of type I muscle fibers were associated with a greater capacity to accumulate distance running at high speeds, as well as a greater capacity to maintain higher output of HSR running during peak periods as duration increases.

Keywords: muscle fiber type composition, fast-twitch, type II fibers, slow-twitch, type I fibers, maximum running velocity, very high-speed running, acceleration, AFL

Quantifying the running performance of Australian football (AF) athletes during match play is important to monitor workloads,[1,2] develop more specific conditioning programs, match tactics, and possibly for recruitment and team selection.[3,4] Nonetheless, understanding and quantifying running performance in AF matches is extremely complex, as a number of athlete- (playing position[1,2]; speed, endurance, and strength[4,5]) and match-related (ie, competition level[5]; match importance and opposition strength[4]) contextual factors may influence these locomotor movements. Indeed, there is substantial between-athletes and between-matches variation in the running performance of AF athletes during matches.[1,2]

It is important to note that most of these contextual factors (playing position, competition level, match importance, and opposition strength) impose an external influence on the running performance of the athlete that cannot be altered. Therefore, they provide limited use in determining what the potential running performance (ie, workload) of an athlete could be. While strength, speed, and endurance offer important information about the potential of an athlete's running performance,[4,5] these physiological characteristics can only be accurately used to determine workload once athletes have developed over the preseason. Alternative measures of an athlete's potential workloads that are not dependent on playing position, opposition status, or training status are required to give coaches the earliest opportunity to understand an athlete's potential running performance and to make decisions about athlete selection and position as well as team tactics.

Muscle fiber typology (MFT) describes the proportion of type I and type II muscle fibers an individual possesses within a given muscle. MFT can differ substantially between individuals and can discriminate between athletes specializing in different sports and disciplines within those sports.[6–8] Individual's with a greater proportion of type II fibers can likely produce more power[9] but may develop more fatigue during maximal intensity exercise.[10,11] Furthermore, possessing a greater proportion of type II fibers may increase the time required to recover from high-intensity exercise, compared with those individuals with a greater proportion of type I fibers.[12] Recent research has suggested that the most elite athletes within a sporting discipline possess a similar MFT,[7,13] where the most accomplished athletes participating in short-duration, highly explosive sports, such as sprinting, weightlifting, and track cycling, tend to possess a greater proportion of type II fibers.[6,7,14] In contrast, athletes participating in longer duration sports/events, often cyclic in nature, such as long-distance runners, triathletes, and road cyclists typically present with a greater proportion of type I fibers.[6,7] Therefore, the determination of MFT may provide useful information about the potential running performance (ie, workloads) of AF athletes.

Bellinger https://orcid.org/0000-0001-7266-2976
Compton https://orcid.org/0000-0002-5818-4450
Bourne https://orcid.org/0000-0002-3374-4669
Derave https://orcid.org/0000-0002-2225-5587
Lievens https://orcid.org/0000-0003-2549-3369
Minahan https://orcid.org/0000-0002-4777-4033
Hopwood (h.hopwood@griffith.edu.au) is corresponding author.

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC

Considering the movement patterns of AF athletes that have been reported in the literature,[15] we hypothesize that the MFT of AF athletes would be heterogenous. Furthermore, given that MFT is associated with peak sprinting power,[9] fatiguability within a sprinting effort, and the time course of recovery following sprinting,[12] it is conceivable that variation in MFT may influence the running performance of AF athletes during matches. Indeed, one study examined the association between MFT and running performance in trained European footballers (ie, soccer).[16] Mohr et al[16] reported that a greater proportion of type II fibers was associated with a significantly greater 1-minute peak and total sprint distance (distance > 6.67 $m \cdot s^{-1}$). Despite these interesting findings, the outcomes were based on one, 90-minute experimental football match played by semiprofessional male football athletes. Few studies have examined the MFT of team-sport athletes, and no studies have reported the MFT of AF athletes. It is likely that the invasive nature of muscle biopsies typically used to determine MFT has prevented the widespread identification of MFT in team-sport athletes.

In 2011, Baguet et al[17] reported a valid method for the noninvasive estimation of MFT using proton magnetic resonance spectroscopy (^{1}H-MRS) to measure carnosine content of the gastrocnemius. This methodology has now been used by the present research group to determine the MFT of numerous athletes across multiple sports.[18,19] Therefore, the purpose of this study was to examine the association between MFT and running performance of professional AF athletes during match play. It was hypothesized that: (1) there would be a large between-athletes variation in MFT and (2) AF athletes with a greater proportion of type II fibers would achieve higher peak running speeds and longer sprinting distances despite displaying a greater decrease in running intensity as duration increases.

Methods

Subjects

This study recruited 24 male professional AF athletes, all currently contracted to a single AF club (mean [SD]; 20.1 [1.9] y, body mass 85.4 [8.0] kg, height 1.89 [0.07] m). After sustaining an anterior cruciate ligament rupture during the preseason period, 1 participant was excluded from the study, leaving 23 participants. Inclusion criteria required that participants (1) were a current listed athlete at an Australian Football League (AFL) club and (2) participated in a minimum of 3 matches of AF, or reserve grade competition. Exclusion criteria required that participants (1) had not consumed beta-alanine 3 months prior to the MFT scans and (2) were not vegan or vegetarian, ensuring the accuracy of the ^{1}H-MRS,[17] due to their modulation of intramuscular carnosine content. Further, data were removed whereby an athlete's performance was severely hindered by injury or otherwise modified. Athletes provided their own written informed consent to participate in this study, which was approved by the Griffith University Human Research Ethics Committee (No. 2019/359).

Design

The present study was a longitudinal observational study. Participants were categorized into their primary playing position as small (n = 10), tall (n = 7), or hybrid (n = 7) groups. The subjects attended a radiology clinic on one occasion to have their soleus and gastrocnemius MFT estimated using ^{1}H-MRS. Global Navigation Satellite System (GNSS) devices quantified the running performance of all relevant match data throughout the 2020 playing season. A total of 224 match files were included, with a mean of 10.4 matches per athlete (range: 4 minimum, 19 maximum). Of included matches, 115 were of reserve-grade standard and 109 AFL standard.

As data collection for this study occurred during the COVID-19 affected 2020 AFL season, the condensed season consisted of reduced match duration (from 20- to 16-min quarters), and teams were required to play more frequently (4-d breaks between matches in some instances). Of the 23 athletes included in the present study, 7 played both AFL and reserve grade and 3 athletes participated in >2 matches of either level.

^{1}H-MRS

Muscle carnosine content was measured by ^{1}H-MRS in the gastrocnemius medialis and soleus muscle of each athletes' right limb to estimate MFT.[17] ^{1}H-MRS measurements were performed on a 3-T whole-body MRI (magnetic resonance imaging) scanner (Philips Medical Systems) as previously described.[18] The carnosine concentration of each muscle was converted to an aggregate z score (ie, CAZ score) relative to an age-matched control population of active, healthy nonathletes, consisting of 40 men.

Global Navigation Satellite System

Catapult Vector devices (S7, Catapult Innovations) sampling at 10 Hz were used for the collection of all locomotive data. The device was placed within a built-in pocket on each athlete's playing jersey, positioned on the upper back, slightly superior to the scapulae, secured via Velcro seal. To minimize interunit error, each athlete was assigned the same unit for the entire season. Raw data (10 Hz) were exported as comma-separated value (.csv) files, downloaded from proprietary software (OpenField, Catapult Innovations) for each training session during preseason and matches. These files include time, velocity, and acceleration. Velocity was adjusted to meters per minute by multiplying by 60. All negative acceleration (in meters per second squared) values (ie, deceleration) were converted to positive values. All files met inclusion criteria of having a horizontal dilution of >1.5.

Various GNSS derived metrics were used to describe match running output. First, velocity-based metrics included were total distance (in meters), high-speed running (ie, HSR distance; >4.17 $m \cdot s^{-1}$), very-high-speed running (ie, VHSR distance > 6.67 $m \cdot s^{-1}$), and maximum velocity ($m \cdot s^{-1}$). These variables (comparable HSR thresholds) have demonstrated acceptable reliability using 10-Hz GNSS devices.[20,21] Additionally, this study included an acceleration-derived GNSS metric. Acceleration (in meters per second squared) was calculated using methods previously explained by Delaney et al,[22] whereby the mean of all rectified acceleration values (in meters per second squared) was determined. This measure has been established as valid[23] and reliable,[20] demonstrating superior reliability to that of traditional acceleration thresholds counts using predefined thresholds (ie, >2.5 $m \cdot s^{-2}$). GNSS-derived metrics are reported as absolute match values due to playing times being comparable between positions: talls: 74 (10.7) minutes, 36 to 104 minutes; hybrid: 71 (9.8) minutes, 36 to 96 minutes; and smalls: 72 (9.8) minutes, 36 to 96 minutes, and across MFT values (median-split MFT groups)—more positive z score: 73 (11) minutes, 36 minimum, 104 maximum; more negative z score: 72 (10) minutes, 36 minimum, 96 maximum.

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC

A moving mean technique was applied to the speed, acceleration, and HSR data to identify the peak intensity of matches across 1- to 10-minute durations for each game file. This technique is used to provide greater estimates of maximal mean running output. GNSS devices sampled at 10 Hz; therefore, a moving mean of 1 minute included 600 data points, while 2 minutes included 1200 data points and so forth. To describe the association between duration and running intensity, a power law relationship ($y = cx^n$)[24,25] was applied to the moving mean data. This method resulted in an intercept (c), representing the highest theoretical intensity as time approaches 0 second, and the slope (n), reflecting the decrement in intensity across time. Intercepts and slopes were calculated for speed, acceleration, and HSR.

Statistical Analysis

All statistical analyses were performed using customized R Studio software (version 1.2.5019, RStudio Inc). The raw carnosine values were used for the statistical analysis, while the CAZ-score values are reported for the descriptive statistics. All data were reported as means (SD) in the text and tables. A significance level of $P < .05$ was used to identify statistical significance across all analyses. Differences in carnosine between playing positions was investigated using t tests (3 separate tests) as there was one sample (data point) per athlete for positional comparisons. Prior to this analysis, data normality was tested using the Shapiro–Wilk test, and the mean carnosine value was normally distributed. Differences between positional groups were described using standardized effect sizes and 95% confidence intervals (CIs), categorized using the thresholds of: 0.2 trivial, 0.21 to 0.60 small, 0.61 to 1.20 moderate, 1.21 to 2.0 large, and 2.0 very large. Analyses were completed using a customized spreadsheet.

To establish the relationship between carnosine values and match physical output measured using total match GNSS values and match intercept and slope values, multiple linear mixed models were fitted. In these models, each of the match-play running performance values (5 variables) and the intercept and slopes were separately included as the outcome variable (dependent variable) and carnosine value as the predictor value (fixed effect). As such, there were a total of 11 separate models constructed (5 GNSS variables and the intercept and slope for speed, high-speed (HS) running, and acceleration). A random intercept and slope design was utilized, where athlete identification was included as a random effect, nested within the athletes playing position. The individual effect of carnosine was accounted for by including the respective value using a random slope structure. While mean carnosine was normally distributed, residual plots (quantile–quantile plots) from each of the fitted models demonstrated normality of the residuals, an assumption of linear mixed models. Associations between MFT and match-play running loads are estimated using the coefficient (c), which describes relationships, due to mean carnosine values being used in analysis, while the SE describes the variation.

Results

The majority of the AF athletes included in the present study had a negative CAZ score (83.3%, –0.60 [0.77]), indicating a greater estimated proportion of type I fibers in most athletes. Despite this, a large range of values were observed across the sample (minimum, maximum: –1.89, 1.25).

Figure 1 displays the distribution of CAZ scores and accompanying carnosine raw values, with AF athletes categorized into playing position. There were trivial to moderate differences observed between positions, where talls had a moderately greater CAZ score than hybrids (effect size = 0.67; CI –0.57 to 2.52) and smalls (effect size = 0.44; CI –0.59 to 2.15). Full results are detailed in Table 1. Despite the moderate effect, a wide CI should be noted.

Linear mixed model parameter estimates assessing the association between MFT and GNSS-derived measures of running performance are presented in Table 2. MFT displayed a significant and negative association with HSR ($c = -207.71$, $P < .001$) and HSR slope ($c = -0.018$, $P < .01$), whereby a greater estimated proportion of type I fibers was associated with an increase in running performance in these metrics. Figure 2 displays the moving mean technique applied to HSR for each athlete. An individual with a higher proportion of type II fibers might be expected to have a greater rate of decline of HSR performance during peak periods of play, as duration increases (x-axis).

Discussion

The purpose of this study was to establish an understanding of the influence MFT has on running performance during professional AF matches. The AF athletes presented with a large range of CAZ scores. Despite this, most athletes possessed a negative CAZ score, indicative of a greater estimated proportion of type I fibers. A greater estimated proportion of type I fibers was associated with a greater accumulation of HSR, while these athletes also demonstrated a greater capacity to maintain HSR performance during peak periods of play (ie, HSR slope), as duration increased.

HSR distance was greater in those AF athletes who possessed a greater estimated proportion of type I fibers. Professional AF athletes cover a mean HSR (>4.17 $m \cdot s^{-1}$) running distance of 2560 (1327) m across the duration of a match. Previous research has demonstrated that playing position influences the volume of distance achieved at HSR intensity (>4.0 $m \cdot s^{-1}$), whereby rucks achieved the lowest distance (2380–2817 m) and midfielders the greatest distance (4166–4462 m). To account for the effect of playing position on the potential association between MFT and running performance, this study included position as a random effect in the statistical models. Further to its association with MFT, the volume of HSR and sprinting (referred to as VHSR) during matches has been shown to markedly differ between elite and subelite AF athletes.[26,27] Interestingly, VHSR was one of the only variables to show a positive association with MFT, along with maximum running velocity and the acceleration intercept. Type II fibers have distinctive mechanical properties, allowing for substantially faster shortening velocities than their type I counterparts.[28] Furthermore, sprint athletes are characterized as having a fast muscle typology compared with their longer distance counterparts.[6,17] One previous study has reported findings on the association between MFT and running performance during an intermittent team sport in European footballers. Mohr et al[16] reported that the ratio of type II to type I fiber proportion of 19 male elite football athletes was positively associated with total and 1-minute peak sprint (>6.67 $m \cdot s^{-1}$) distance during an experimental football match and that the ratio of type II to type I fiber proportion explained 69% of the variance in sprint distance (comparable to VHSR in the present study) achieved during match play. In the present study, although no significant association was determined with MFT and VHSR ($c = 12.86$, $P = .418$), it was one of the only variables that showed a positive association with the CAZ score, whereby the direction of the association indicated that a greater proportion of type II fibers was associated with a greater accumulation of VHSR.

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC

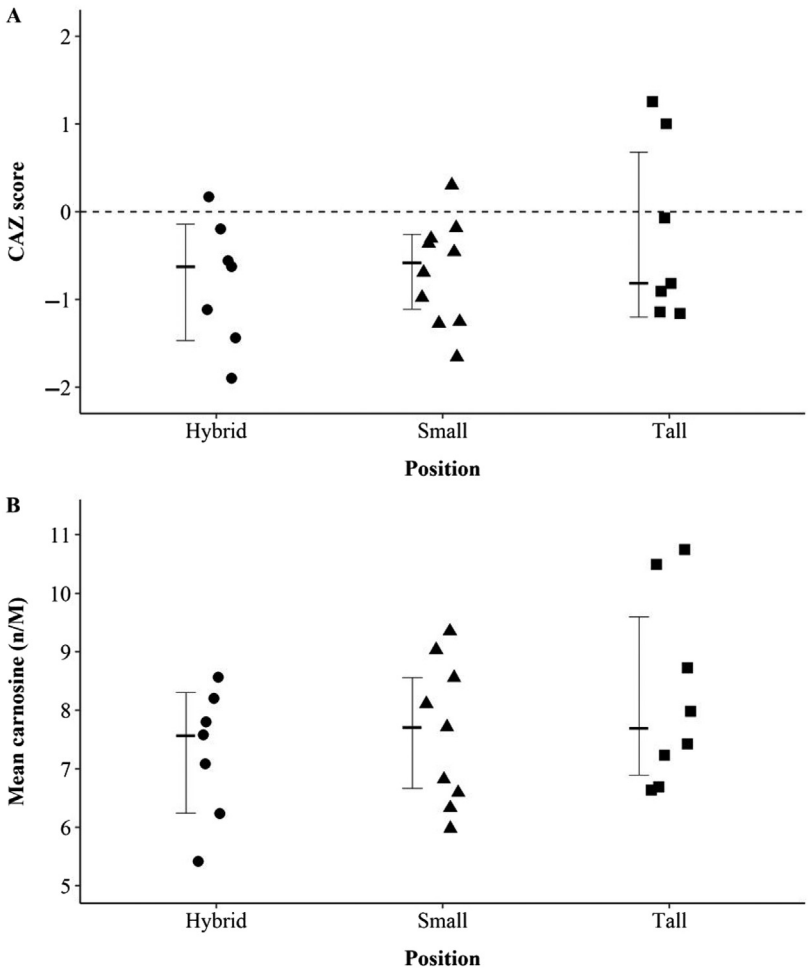

Figure 1 — CAZ score and mean carnosine concentration (in millimoles) distributions between hybrid (n = 7; circles), small (n = 10; triangles), and tall (n = 7; squares) playing positions, with median and 95% CIs. CAZ score indicates carnosine aggregate *z* score.

Table 1 Difference in Muscle Fiber Typology Between Positions

	Effect size	95% CI	*P*
Small – hybrid	0.29	−1.61 to 0.93	.57
Tall – hybrid	0.67	−0.57 to 2.52	.20
Tall – small	0.44	−0.59 to 2.15	.39

Table 2 Relationship Between CAZ Score and Variables of Match-Play Movement Profiles

Variable	Coefficient (*c*)	SE	*P*
Maximum velocity, $m \cdot s^{-1}$	0.35	0.20	.104
Total distance, m	−17.54	98.41	.862
High-speed running, m	−207.71	43.50	<.001***
Very high-speed running, m	12.86	14.36	.418
Acceleration, $m \cdot s^{-2}$	−0.005	9.52	.365
Speed intercept, $m \cdot min^{-1}$	−4.42	1.99	.066
Speed slope, $m \cdot min^{-1}$	−0.003	0.002	.069
High-speed intercept, $m \cdot min^{-1}$	−1.44	2.53	.589
High-speed slope, $m \cdot min^{-1}$	−0.018	0.005	.009**
Acceleration intercept, $m \cdot s^{-2}$	0.004	0.009	.695
Acceleration slope, $m \cdot s^{-2}$	−0.0002	0.002	.912

**$P < .01$. ***$P < .001$.

Both of these results suggest that MFT has some influence on an athlete's high-intensity running performance. Despite these compelling preliminary findings, more research is required to fully elucidate the deterministic potential of MFT on GNSS-derived metrics of running output in AF.

The present study demonstrated a significant negative association between CAZ score and HSR slope, whereby AF athletes with a greater estimated proportion of type I fibers were able to better maintain performance of HSR across 1 to 10-minute peak periods of match play. These findings may be underpinned by the superior fatigue resistance of type I compared with type II fibers. AF athletes have been reported to complete approximately 2 high-speed (>3.9 $m \cdot min^{-1}$) efforts per minute,[26] 22 (9) sprints (≥7 $m \cdot s^{-1}$) over the course of a match, with 45% of these occurring with less than 30 seconds of recovery.[15] Further to this, AF athletes typically perform 82 (26) acceleration efforts during match play (threshold of ≥2.78 $m \cdot s^{-2}$),[15] which incur a very high metabolic and neuromuscular cost.[29] It is clear that AF requires athletes to possess a large capacity to tolerate extremely high running demands. Previous research has demonstrated that recreational individuals

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC

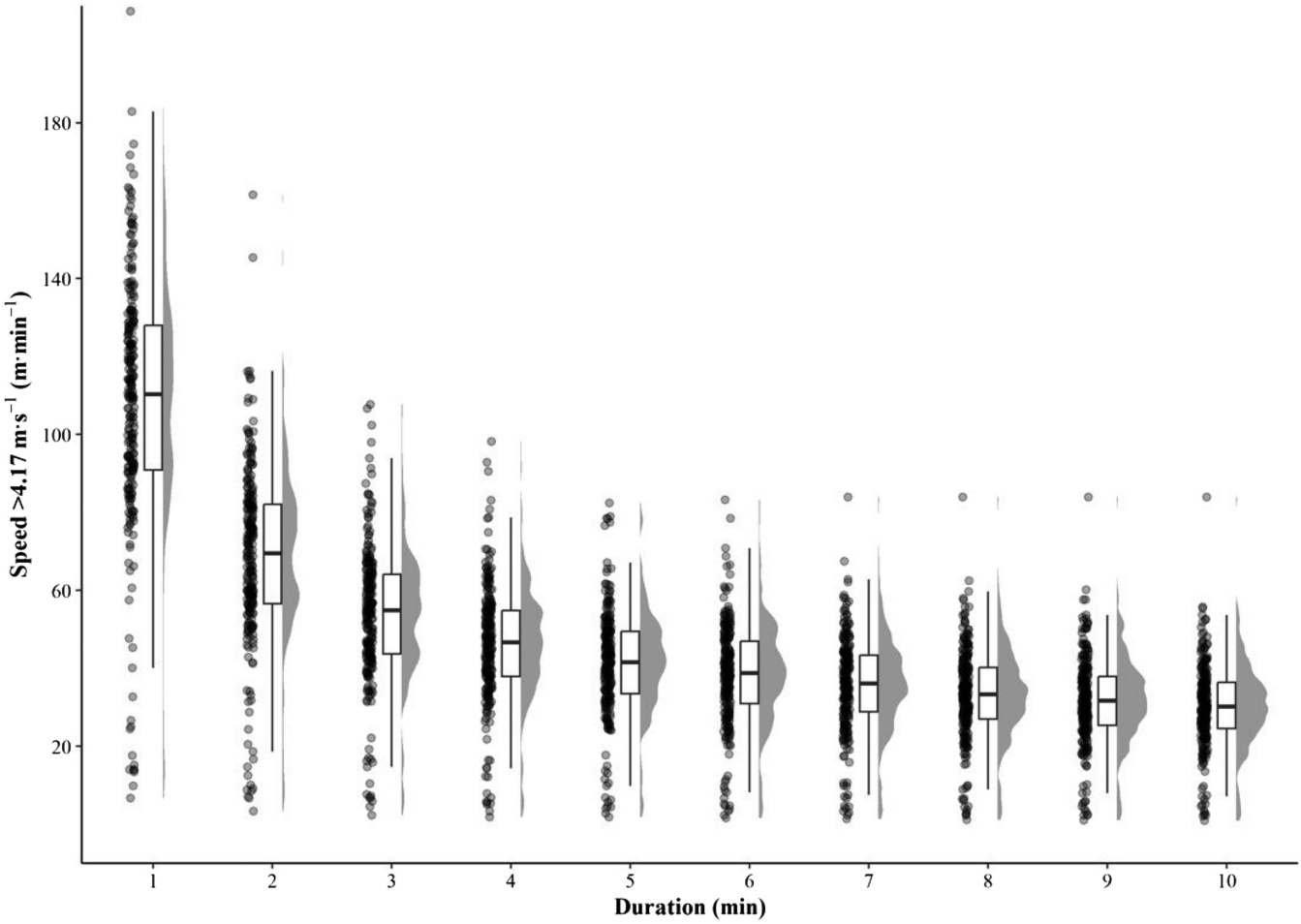

Figure 2 — Maximal mean high-speed running >4.17 m·s^{-1} (in meters per minute) by duration during Australian Rules match play. Data are presented as individual responses, box plots, and raincloud plots demonstrating the distribution of data.

with a greater estimated proportion of type II fibers had a greater degree of performance decrement both within and between maximal 30-second cycling sprints, as well as a longer time course of recovery following the sprint efforts.[12] In light of these findings, AF athletes with a greater estimated proportion of type II fibers may not be able to tolerate repeated exposures to high-intensity actions, which is evidenced by a greater HSR slope. Future experimental studies should explore whether tactical substitutions can mitigate the magnitude of the decline in the speed slope in AF athletes with a greater estimated proportion of type II fibers.

A large interpositional and intrapositional range of MFT was exhibited within professional AF athletes, suggesting there is not a typical MFT in professional AF athletes. Similar interpositional heterogeneity was reported by Lievens et al,[19] whereby the MFT of the soleus quantified by the same method employed in the present study did not differ between playing positions in Belgian Jupiler Pro League or Premier League European footballers (range: −1.85 to 2.22).[19] Furthermore, 84.7% of players had a soleus z score of <0.5, which is comparable to the present study (91.7%). These findings along with those of the present study suggest that position-specific MFT for AF and European football is unlikely to exist, yet the majority of players possess intermediate (CAZ score between −0.5 and 0.5) or slow-typology (CAZ score less than or equal to −0.5) muscle profiles.[19] MFT is a distinguishing characteristic for event specialization and training status in sprint and endurance disciplines in many racing sports (ie, track running and professional cycling), while differences have been noted in elite middle-distance runners.[17] Despite there being no distinctive CAZ score in the present study, akin to previous research in European soccer athletes,[19] there was a large affinity to negative CAZ scores, indicating a greater estimated proportion of type I fibers. AF has become a highly endurance-based sport, whereby athletes cover distances of 12,897 (1601) m per match. Provided this information and the effect MFT has on recoverability, it is reasonable to question whether athletes with fast typology (ie, CAZ score ≥ 0.5) would have the capacity to have a major impact at the highest level given the substantial endurance demands during an AF match. Nonetheless, athletes with different physiological profiles can be balanced within team sports due to tactics and specialist positions which may be reflective of only a small population within the sample. This might be most true for team sports such as rugby union, whereby substitutions are planned based on the period of the match; large, heavier players on field at the beginning with burst (ie, high intensity and high fatigue) players often substituted on field toward the end of a match, where they are afforded the luxury of covering much less running distance. It has been demonstrated in the literature that in rugby union the substituted player often achieves greater distances, and high-intensity running actions

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC

compared with starting players.[30] It is likely that a team might be balanced with different athletic qualities (ie, elite endurance specialists, or pure strength and speed in others) in different athletes, to provide a tactical advantage on competition.

Indeed, several contextual influencing factors impact an AF athlete's running performance. It should be noted as limitations of this study that data were collected during a modified COVID-19 impacted season, particularly when comparing data with previous research. Further, despite small differences, data from 2 levels of competition were utilized to increase the study sample, noting that peak intensities were of primary interest.

Conclusion

The findings from this study suggest that there is not a typical muscle fiber typology (MFT) best suited to the sport of Australian football (AF) or a position-specific MFT within AF. However, most athletes had negative CAZ-score values, which is indicative of a greater estimated proportion of type I fibers. A greater estimated proportion of type I fibers was associated with greater high-speed-running distance during match play and an enhanced capacity to maintain high-speed running during peak periods of increasing duration. These findings highlight the importance of MFT in relation to key GNSS-derived metrics of AF match play.

Practical Applications

- Athletes with a greater proportion of type I muscle fibers may be able to cover more distance at high velocities during Australian football (AF).
- Athletes with a greater proportion of type I muscle fibers may have an enhanced capacity to maintain high-speed-running performance during peak periods of AF match play with longer durations.
- Tactical substitutions may mitigate the magnitude of the decline in the speed slope in AF athletes with a greater estimated proportion of type II fibers.

References

1. Coutts AJ, Kempton T, Sullivan C, Bilsborough J, Cordy J, Rampinini E. Metabolic power and energetic costs of professional Australian football match-play. *J Sci Med Sport*. 2015;18(2):219–224. doi:10.1016/j.jsams.2014.02.003
2. Delaney JA, Thornton HR, Burgess DJ, Dascombe BJ, Duthie GM. Duration-specific running intensities of Australian football match-play. *J Sci Med Sport*. 2017;20(7):689–694. doi:10.1016/j.jsams.2016.11.009
3. Pyne DB, Gardner AS, Sheehan K, Hopkins WG. Fitness testing and career progression in AFL football. *J Sci Med Sport*. 2005;8(3):321–332. doi:10.1016/s1440-2440(05)80043-x
4. Ryan S, Coutts AJ, Hocking J, Kempton T. Factors affecting match running performance in professional Australian football. *Int J Sports Physiol Perform*. 2017;12(9):1199–1204. doi:10.1123/ijspp.2016-0586
5. Johnston RD, Black GM, Harrison PW, Murray NB, Austin DJ. Applied sport science of Australian football: a systematic review. *Sports Med*. 2018;48(7):1673–1694. doi:10.1007/s40279-018-0919-z
6. Costill DL, Daniels J, Evans W, Fink W, Krahenbuhl G, Saltin B. Skeletal muscle enzymes and fiber composition in male and female track athletes. *J Appl Physiol*. 1976;40(2):149–154. doi:10.1152/jappl.1976.40.2.149
7. Bex T, Baguet A, Achten E, Aerts P, De Clercq D, Derave W. Cyclic movement frequency is associated with muscle typology in athletes. *Scand J Med Sci Sports*. 2017;27(2):223–229. doi:10.1111/sms.12648
8. Lievens E, Bellinger P, Van Vossel K, et al. Muscle typology of world-class cyclists across various disciplines and events. *Med Sci Sports Exerc*. 2021;53(4):816–824. doi:10.1249/MSS.0000000000002518
9. Bellinger P, Bourne MN, Duhig S, et al. Relationships between lower limb muscle characteristics and force-velocity profiles derived during sprinting and jumping. *Med Sci Sports Exerc*. 2021;53(7):1400–1411. doi:10.1249/MSS.0000000000002605
10. Aagaard P, Andersen JL. Correlation between contractile strength and myosin heavy chain isoform composition in human skeletal muscle. *Med Sci Sports Exerc*. 1998;30(8):1217–1222. doi:10.1097/00005768-199808000-00006
11. Tihanyi J, Apor P, Fekete G. Force-velocity-power characteristics and fiber composition in human knee extensor muscles. *Eur J Appl Physiol*. 1982;48(3):331–343. doi:10.1007/BF00430223
12. Lievens E, Klass M, Bex T, Derave W. Muscle fiber typology substantially influences time to recover from high-intensity exercise. *J Appl Physiol*. 2020;128:648–659. doi:10.1152/japplphysiol.00636.2019
13. Bellinger P, Lievens E, Kennedy B, Rice H, Derave W, Minahan C. The muscle typology of elite and world-class swimmers. *Int J Sports Physiol Perform*. 2022;17:1179–1186. doi:10.1123/ijspp.2022-0048
14. Serrano N, Colenso-Semple LM, Lazauskus KK, et al. Extraordinary fast-twitch fiber abundance in elite weightlifters. *PLoS One*. 2019;14(3):e0207975. doi:10.1371/journal.pone.0207975
15. Varley MC, Gabbett T, Aughey RJ. Activity profiles of professional soccer, rugby league and Australian football match play. *J Sports Sci*. 2014;32(20):1858–1866. doi:10.1080/02640414.2013.823227
16. Mohr M, Thomassen M, Girard O, Racinais S, Nybo L. Muscle variables of importance for physiological performance in competitive football. *Eur J Appl Physiol*. 2016;116(2):251–262. doi:10.1007/s00421-015-3274-x
17. Baguet A, Everaert I, Hespel P, Petrovic M, Achten E, Derave W. A new method for non-invasive estimation of human muscle fiber type composition. *PLoS One*. 2011;6(7):e21956. doi:10.1371/journal.pone.0021956
18. Bellinger P, Desbrow B, Derave W, et al. Muscle fiber typology is associated with the incidence of overreaching in response to overload training. *J Appl Physiol*. 2020;129(4):823–836. doi:10.1152/japplphysiol.00314.2020
19. Lievens E, Van-Vossel K, Van-De-Casteele F, et al. Muscle fibre typology as a novel risk factor for hamstring strain injuries in professional football (soccer): a prospective cohort study. *Sports Med*. 2021;52(1):177–185. doi:10.1007/s40279-021-01538-2
20. Thornton HR, Nelson AR, Delaney JA, Duthie GM. Inter-unit reliability and effect of data processing methods of global positioning systems. *Int J Sports Physiol Perform*. 2019;14(4):432–438.
21. Johnston RJ, Watsford ML, Kelly SJ, Pine MJ, Spurrs RW. Validity and interunit reliability of 10 Hz and 15 Hz GPS units for assessing athlete movement demands. *J Strength Cond Res*. 2014;28(6):1649–1655. doi:10.1519/JSC.0000000000000323
22. Delaney JA, Duthie GM, Thornton HR, Scott TJ, Gay D, Dascombe BJ. Acceleration-based running intensities of professional rugby league match play. *Int J Sports Physiol Perform*. 2016;11(6):802–809. doi:10.1123/ijspp.2015-0424
23. Delaney JA, Wileman TM, Perry NJ, Thornton HR, Moresi MP, Duthie GM. The validity of a global navigation satellite system for quantifying small-area team-sport movements. *J Strength Cond Res*. 2019;33(6):1463–1466. doi:10.1519/JSC.0000000000003157

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC

24. Delaney JA, Thornton HR, Rowell AE, Dascombe BJ, Aughey RJ, Duthie GM. Modelling the decrement in running intensity within professional soccer players. *Sci Med Footb*. 2017;2(2):86–92. doi:10.1080/24733938.2017.1383623
25. Katz L, Katz JS. Fractal (power law) analysis of athletic performance. *Sports Sci Med Rehab*. 1994;5(2):95–105. doi:10.1080/15438629409512005
26. Johnston RJ, Watsford ML, Austin DJ, Pine MJ, Spurrs RW. Movement demands and metabolic power comparisons between elite and subelite Australian footballers. *J Strength Cond Res*. 2015; 29(10):2738–2744. doi:10.1519/JSC.0000000000000936
27. Brewer C, Dawson B, Heasman J, Stewart G, Cormack S. Movement pattern comparisons in elite (AFL) and sub-elite (WAFL) Australian football games using GPS. *J Sci Med Sport*. 2010;13(6):618–623. doi:10.1016/j.jsams.2010.01.005
28. Larsson L, Moss RL. Maximum velocity of shortening in relation to myosin isoform composition in single fibres from human skeletal muscles. *J Physiol*. 1993;472:595–614. doi:10.1113/jphysiol.1993.sp019964
29. Hader K, Mendez-Villanueva A, Palazzi D, Ahmaidi S, Buchheit M. Metabolic power requirement of change of direction speed in young soccer players: not all is what it seems. *PLoS One*. 2016;11(3): e0149839. doi:10.1371/journal.pone.0149839
30. Lacome M, Piscione J, Hager JP, Carling C. Analysis of running and technical performance in substitute players in international male rugby union competition. *Int J Sports Physiol Perform*. 2016;11(6): 783–792. doi:10.1123/ijspp.2015-0191

Authenticated t_lambrinides@yahoo.com | Downloaded 10/12/23 12:59 PM UTC